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MILITARY SPECIFICATION

CONTROL ENCLOSURE CY-6816/APX-72 FOR TRANSPONDER SET CONTROL C-6280( ) (P)/APX

This specification is approved for use by the Naval Electronic Systems Command.

1. SCOPE

1.1 This specification covers the design, fabrication, and tests of the CY-6816/APX Control Enclosure. The purpose of this enclosure is to permit mounting of the C-6280/APX-72 Control box aboard Naval vessels.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (HIGH IMPACT), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-Q-9858	- Quality Program Requirements
MIL-P-15024	- Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-E-16400	- Electronic Equipment, Naval Ships and Shore, General Specification (Navy)
MIL-E-17555	- Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-108	- Definition of and Basic Requirements for Enclosure for Electric and Electronic Equipment

FSC 5840

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MIL-STD-167 - Mechanical Vibration of Shipboard Equipment
MIL-STD-454 - Standard General Requirements for Electronic Equipment

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of his specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402)

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with all provisions of MIL-E-16400, MIL-STD-108 and MIL-STD-454, except as hereinafter specified. Screws and screw thread devices shall be in accordance with Handbook H28.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3)

3.3 Simplification and improvements. The provisions of this specification shall not preclude simplification or improvement, except that any changes of this nature shall be approved by the command or agency concerned.

3.4 Environment. The assembled equipment shall be capable of satisfactory operation under the following conditions:

(a) Temperature - Class 2
(b) Shock - Grade A, Class I, lightweight, type A of MIL-S-901
(c) Vibration - Type I of MIL-STD-167
(d) Humidity - Continuous or intermittent, up to 95 percent relative humidity
(e) Enclosure - Dripproof, per MIL-STD-108

3.5 Primary power line voltage and frequency. Primary power line voltage and frequency is not required.

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3.6 Synchros. Synchros are not required.

3.7 Controls and indicators. The front panel of the enclosure shall be provided with a Mode 4 Caution Light and test switch for this light. 1/ The front panel shall be labeled as shown in Figure 1 for this light and switch. The switch shall be a single pole, single throw type. In the NORMAL position the switch shall be open. In the TEST position, the switch shall be spring-loaded so as to permit contact closure only when the switch is depressed. The switch and light are to be connected to the terminal board as shown in Figure 2.

3.8 Size. The unit shall be no larger than 8 inches wide, 10 inches high by 8 inches deep.

3.9 Time meters. Time meters are not required.

3.10 Construction details. The unit shall be designed to mount the C-6280 Transponder Set Control in a dripproof case in such a manner that the C-6280 shall be readily removable from the front panel of the case. The front panel shall also be removable from the case without removing the C-6280.

3.10.1 Mounting. Provision shall be made for mounting the CY-6816 on its rear surface only.

3.10.2 Terminal boards. Three twenty terminal boards shall be provided, mounted in the rear of the interior of the CY-6816 case. The terminal boards and terminals shall be labeled in the rear of the case adjacent to the terminal boards as designated in Figure 2.

3.10.3 Connector cable harness. The contractor shall provide the mating connector (MS24266-R-22B-55S) to the C-6280 together with a harness to the terminal boards of sufficient length so as to permit removal of the C-6280 from the CY-6816 case without disconnection of the C-6280 connector. The mating connector shall be connected to the terminal boards using the pin numbers and terminal numbers as shown in Figure 2.

3.10.4 Cable entrances. The design of the CY-6816 shall be such as to permit cable entrance on the top, bottom, or either side of the case, by means of stuffing tubes or the like. When requested by the contractor, the Navy shall provide information regarding these cable entrances so that the contractor may then provide information and direction for the provision of these cable entrances by the installing activity. The type and manner of entrance shall be provided by the design agency, who shall then approve these directions.

1/ To be operated on 28V d.c.

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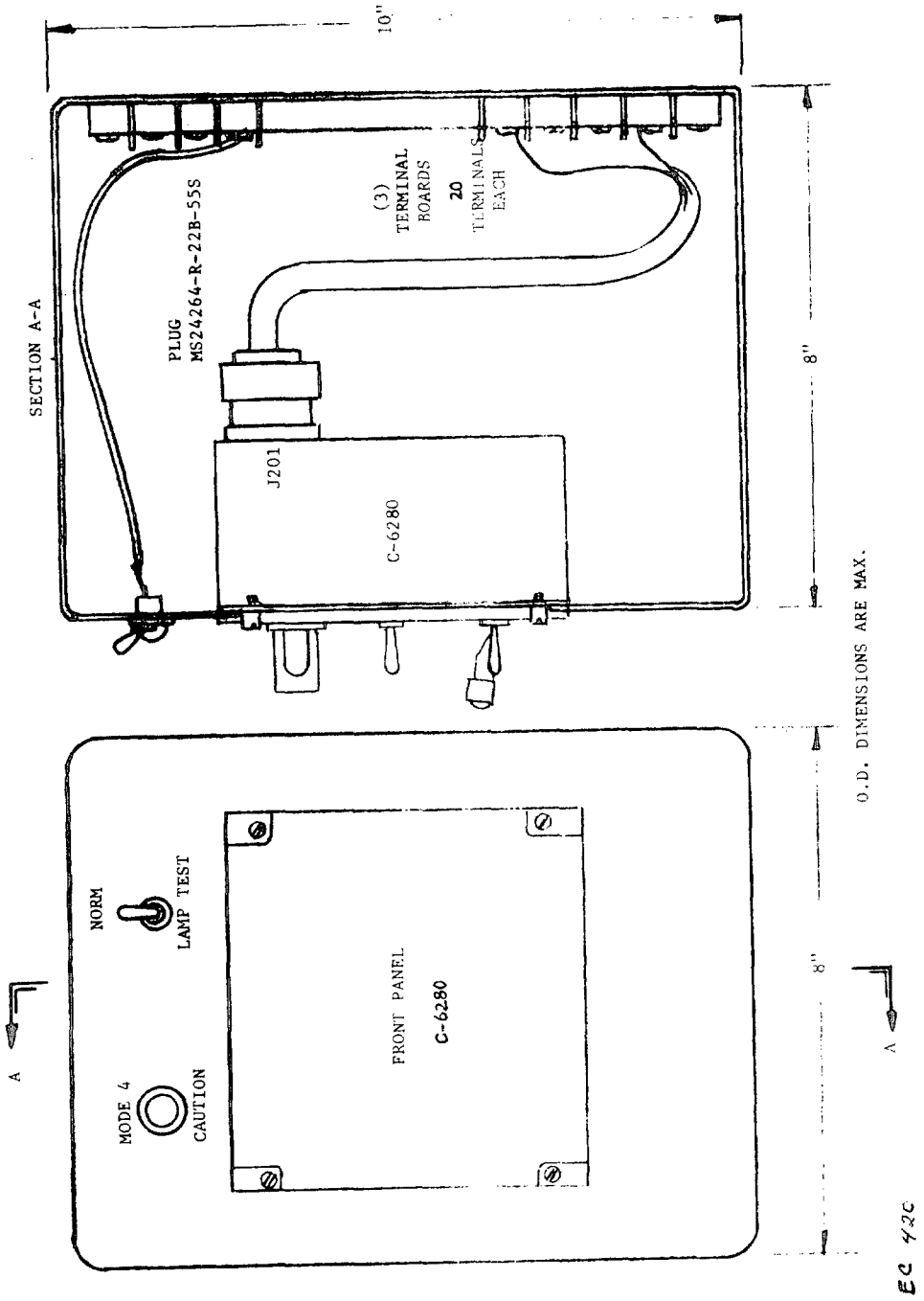


FIGURE 1. C-6280 enclosure

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3.10.5 Wiring directions. The contractor shall provide a photoetch metal plate showing all wiring hook-up including the terminal board jumpers as shown in Figure 2. The metal plate shall be photoetch black letters on white background in accordance with Type A of MIL-P-15024. This plate shall be mounted within the C-6816 case.

3.11 Blown fuse indicators. Blown fuse indicators are not required.

3.12 Instruction plates. Instruction plates are not required.

3.13 Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility. Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility is not required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered product to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Visual and mechanical examination (see 4.5.1)

(c) Inspection of preparation for delivery (see 4.6)

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4.3 First article inspection. Unless otherwise specified (see 6.2.1) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection

Examination or test	Test method paragraph	Number of sample units to be inspected	Number of failures
Visual and mechanical examination	4.5.1	all	0
Operating test	4.5.2	all	0
Dimensions	4.5.3	all	0
Shock and vibration	4.5.5	all	0
Salt spray test	4.5.6	all	0

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every enclosure offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and tests which detect deviations from design and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the requirements of table II.

TABLE II. Production inspection

Examination or test	Test method paragraph
Visual and mechanical examination	4.5.1
Operating test	4.5.2

4.5 Inspection procedures.

4.5.1 Visual and mechanical examination. Each enclosure shall be examined for workmanship criteria listed in requirement 9 of MIL-STD-454. Particular attention shall be given the following:

(a) Workmanship, assembly and fit, and marking
(b) Materials, parts and finish
(c) Treatment for prevention of corrosion
(d) Mounting provisions
(e) Insulation protection

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4.5.2 Operating test. Tests for illuminating the Mode 4 caution light in the TEST position shall be made by operation of the front panel switch to verify compliance with 3.7.

4.5.3 Dimensions. The external dimensions of the enclosure shall be measured to determine conformance to 3.8. Location of mounting holes for the C-6280 Transponder Set Control shall be measured or gaged to determine conformance to 3.10.

4.5.4 Enclosure test. The enclosure shall be visually examined in accordance with MIL-STD-108 (dripproof) to determine the ability of the enclosure to exclude falling drops of liquid or solid particles and to verify conformance to 3.4 (e).

4.5.5 Shock and vibration. The enclosure, with the C-6280 Transponder Set Control installed, shall be shock and vibration tested in accordance with MIL-E-16400 to determine compliance with 3.4 (b) and 3.4 (c).

4.5.6 Salt spray test. The salt spray test shall be made in accordance with MIL-E-16400. At the conclusion of the test the enclosure shall be carefully examined in detail to detect evidence of physical degradation such as deterioration or corrosion of any enclosure surface or component which could in any manner prevent the enclosure from meeting operational requirements during service life.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order).

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use in the mounting of the C-6280/APX-72 Control box aboard naval vessels.

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6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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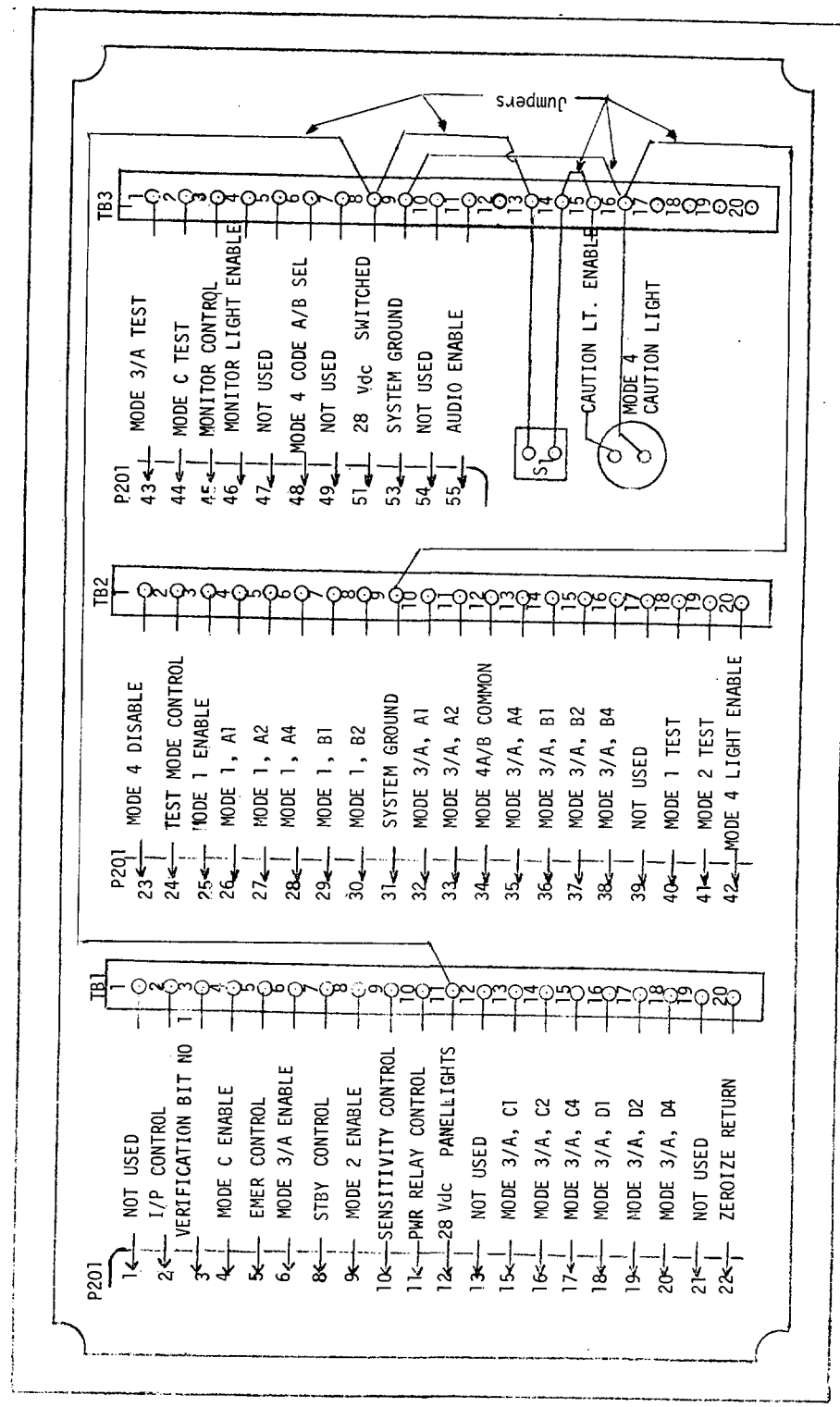


FIGURE 2. CY-6816 WIRING DIAGRAM

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